

Temperature Fragility in Instruction-Tuned Language Models

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Abstract

Truncation samplers such as top- p , min- p , and top- $n\sigma$ exist to prevent the accuracy collapse of high-temperature sampling, and their evidence comes from temperatures where every model has already collapsed. We ask how instruction-tuned models behave at the temperatures deployments use, from the default temperatures of 0.6 to 1.0 up to a test temperature of $T=1.3$ above that range, and when a truncation rule changes the outcome. In one controlled pipeline we run nine models from five families across eight decoding configurations, four quantization levels, and two tasks with a checkable answer, GSM8K and MMLU-Pro. Temperature sensitivity varies sharply by model. Under plain temperature sampling at $T=1.3$, Llama-3.1-8B and Llama-3.2-3B lose 30 to 38 accuracy points and Llama-3-8B loses 17 on the longer task, while no model outside the family loses more than 10. The loss is degeneration. Generations become incoherent, run on until the maximum token count, and never reach an answer, while the well-formed answers stay as accurate as under greedy decoding. The collapse replicates on both Llama checkpoints through a second inference engine, at native precision for Llama-3.2-3B, so neither the engine nor the quantization causes it. At the default temperatures no truncation rule gains more than 3 points on any robust model (8 at the 95% upper limit); gains of 3 to 10 points appear only at $T=1.3$ on the models nearest their own collapse temperature, and tests up to $T=2.0$ show each model collapsing at a different temperature, with truncation improving accuracy only above it.

1 Introduction

Sampling is the default decoding choice for deployed language models (Shi et al., 2024), and truncation samplers exist to prevent accuracy loss at high temperature. They were designed for open-ended generation, where output variety is the goal

(Garces Arias et al., 2024), but they are now also evaluated on tasks with one checkable answer. Min- p reports accuracy gains over top- p on GSM8K at high temperature (Nguyen et al., 2025), and top- $n\sigma$ reports accuracy maintained up to $T=3.0$ (Tang et al., 2024). Both results assume that raising the temperature eventually reduces every model’s accuracy and a good sampler reduces that loss. We test that premise at the temperatures deployments actually use and ask **which models collapse in that range, and whether the sampler changes anything before they do.**

There is a good reason to expect the collapse. At each step the decoder draws the next token at random from the model’s predicted distribution, and a temperature parameter scales the randomness of the draw (greedy decoding instead always takes the most probable token). Raising it moves probability from the model’s preferred tokens toward the tail of the distribution, the tokens with low predicted probability, where the model’s estimates are least reliable, so a high-temperature draw can select a token that fits the context poorly (Holtzman et al., 2020; Hewitt et al., 2022). The selected token then becomes context for every later step, and the continuation can degenerate into incoherent text. Truncation samplers prevent this degeneration by deleting the tail of the distribution before the draw, and the rules differ in which tokens they exclude.

The accuracy evidence has two problems. First, it generalizes from very few models. Top- $n\sigma$ ’s results come from a single model, Llama-3-8B (Tang et al., 2024), and min- p ’s main results from Mistral-7B (Nguyen et al., 2025). Second, the gains are demonstrated at temperatures of 1.5 to 3, while deployed systems set defaults between 0.6 and 1.0 (Section 3).

Across nine instruction-tuned models from five families, run through one controlled pipeline, temperature sensitivity inside that range turns out to be a property of the model. Six models lose no more

than 10 points at $T \leq 1.3$ under every sampler we test. Llama-3.1-8B and Llama-3.2-3B lose 30 to 38 points under plain temperature sampling, and Llama-3-8B, top- $n\sigma$'s test model, loses 17 on the longer task.

Our contributions:

- **Heterogeneous temperature sensitivity, measured in one pipeline.** Seven instruction-tuned models, four GGUF quantization levels, eight decoding configurations, and $T \in \{0.7, 1.0, 1.3\}$ on GSM8K and MMLU-Pro ($N=50$ items, $K=3$ repetitions), 185,600 generations, plus the same temperature tests on two further Llama releases, with item-clustered bootstrap CIs on every main estimate. Every model that loses more than 10 points is a Llama-3 release (Section 4.1).
- **A replicated failure pattern.** During the collapse, generations become incoherent and run on until the token limit. Among Llama's generations that reach a well-formed answer at $T=1.3$, accuracy matches greedy decoding on the same items, so the lost generations are the ones that never reach an answer (Section 4.3). The same failure occurs under a second engine and at native precision: through HF transformers, Llama-3.2-3B collapses the same way in native BF16 and Llama-3.1-8B in 8-bit, while Qwen3-4B stays flat (Section 4.2).
- **A conditional sampler benefit.** At the default temperatures no truncation rule gains more than 3 points on any robust model (8 at the 95% upper limit). Gains appear only as a model approaches its cliff, the lowest temperature at which it loses more than half its accuracy, and tests up to $T=2.0$ put the three cliffs we measure at different temperatures, Llama's inside the deployment range (Section 4.4).

2 Related work

Truncation samplers. Top- k sampling draws from only the k most probable tokens; it was introduced for story generation, where likelihood-maximizing decoding produced generic and repetitive text (Fan et al., 2018). Holtzman et al. (2020) showed that a fixed k fits poorly because the next-token distribution is sometimes peaked and sometimes flat, and proposed nucleus sampling, which keeps the smallest set of tokens whose probabilities sum to a target p . Min- p keeps only the to-

kens whose probability is at least a set fraction of the highest-probability token's, so the rule excludes more tokens when that probability is high and fewer when it is low (Nguyen et al., 2025). Top- $n\sigma$ selects tokens using logits, the scores the model assigns before converting them to probabilities, keeping tokens within n standard deviations of the top logit; temperature rescales all logits equally, so the retained set does not change with temperature (Tang et al., 2024). Welleck et al. (2020) proved that top- k and nucleus decoding can produce sequences that never terminate even when the model itself always terminates, so non-termination is a documented decoding failure. In contrast to the sampler papers, which compare truncation rules against each other, we test whether the failure they all aim to prevent occurs at deployment temperatures at all.

Reproducibility of sampler gains. Schaeffer et al. (2025) reanalyzed min- p on nine models and 31 temperatures, giving every sampler an equal number of hyperparameter settings, and found its reported gains "largely indistinguishable" from the other samplers'. A second variable in such a comparison is the order in which temperature and truncation are applied. The order matters because the truncation threshold is computed on whichever distribution the rule sees: applied before temperature, the rule cuts the model's original, sharp distribution; applied after, it cuts one that temperature has already flattened, so the same nominal setting keeps a different set of tokens. We run the two chain orders as explicit configurations (Section 3). Garces Arias et al. (2024) tests a range of decoding parameter values on open-ended generation with quality and diversity metrics, while our tasks have one checkable answer. Shi et al. (2024) compare decoding methods on closed-ended tasks and report that the best method changes with the task and the model, and that quantization alters how robust a model is to the choice; the comparison is again among methods, and it does not establish whether models lose accuracy at deployment temperatures. Song et al. (2025) documents per-model gaps between greedy decoding and sampling on reasoning benchmarks and treats the gaps as evaluation noise to report; we study the difference and find that it results from a termination failure in one model family.

Robustness to decoding randomness. Renze and Guven (2024) sampled temperatures from 0.0

to 1.6 on multiple-choice problem solving and found no statistically significant effect on accuracy between 0.0 and 1.0. We also find no statistically significant effect for most models, but per-model results show Llama-3.1-8B losing accuracy just above $T=1.0$ while the other models remain comparatively stable (Section 4.1). The nearest study is Grover (2026), which also asks what makes a model robust to decoding randomness and concludes that instruction tuning is the primary determinant. Our data refine that conclusion. All nine of our models are instruction-tuned and one family still collapses, so instruction tuning alone does not explain the differences. Their evaluation differs from ours in two respects that matter. They test temperatures only up to $T=1.0$, and the Llama collapse happens between 1.0 and 1.3; and their robust models score 11 to 30% on synthetic tasks, so there is little accuracy left to lose. They also test no truncation samplers. Fastowski et al. (2025) connect token-distribution entropy to temperature sensitivity on factual question answering; our analysis of token scores supports this association (Appendix K), though model size does not determine temperature sensitivity in our experiments, where Qwen3-1.7B maintains its accuracy and the larger Llama-3.1-8B collapses.

3 Experimental setup

The study tests every combination of model, quantization level, decoding configuration, and temperature on two tasks. Table 1 lists the grid and the additional experiments. We call one model at one quantization level, on one task at one temperature, a *cell*; we test the eight decoding configurations in every cell on the same fixed set of 50 items. Every sampler comparison in the paper therefore keeps the cell fixed and changes the decoding configuration, and every temperature comparison keeps the configuration fixed and changes only the temperature. We keep the other experimental settings fixed: one llama.cpp commit, each model’s official chat template rendered client-side, the native completion API, and per-record seeds derived deterministically from the cell and configuration. An accuracy difference between two runs is then attributable to the experimental factors, up to random variation from sampling that the bootstrap below quantifies.

Decoding. We test eight decoding configurations in each cell: a deterministic greedy baseline,

plain temperature sampling, and the four truncation rules of Section 2, each at the setting its source paper uses or recommends (top- p 0.95 (Holtzman et al., 2020), top- k 40 (Radford et al., 2019), min- p 0.05 (Nguyen et al., 2025), top- $n\sigma$ 1.0 (Tang et al., 2024)). Because the order of temperature and truncation changes which tokens remain (Section 2), we apply temperature scaling before truncation in the truncation configurations and add two ablation configurations that apply it after (the llama.cpp default) for top- p and min- p , the two rules whose retained set depends on that order. Temperatures $T \in \{0.7, 1.0, 1.3\}$ span the deployment band. The default temperatures that models and engines set lie between 0.6 and 1.0 (Llama-3.1’s generation config sets 0.6, llama.cpp defaults to 0.8, the OpenAI and Anthropic APIs to 1.0),¹ so 0.7 and 1.0 represent common default temperatures and 1.3 lies just above them. Every configuration that samples tokens runs $K=3$ repetitions per item; greedy decoding is deterministic, so $K=1$.

Tasks. We evaluate GSM8K, grade-school math word problems (Cobbe et al., 2021) (0-shot chain-of-thought with the prompt of Meta’s Llama evaluation suite (Meta AI, 2024c), numeric exact match), and MMLU-Pro, ten-option multiple-choice questions across 14 categories (Wang et al., 2024) (0-shot chain-of-thought, letter exact match), $N=50$ items each, a 512-token budget on GSM8K and 1024 on MMLU-Pro (the cap), with parsers faithful to lm-eval-harness (Gao et al., 2023). Both tasks require a chain of reasoning ending in one exactly checkable answer, which avoids variation from subjective grading and leaves a long generation in which coherence can be lost. The two tasks differ in generation length, with MMLU-Pro’s generated reasoning containing roughly twice as many tokens as GSM8K’s, a difference Section 4.4 uses to test whether longer generations fail earlier. Both tasks grade a generation by parsing a final answer out of the generated text, so the parser decides how a generation that never reaches an answer is graded. We log the parse path of every record: strict when the specified final-answer sentence is present (“The final answer is 26”; “The answer is (I)”), flexible when it is absent and the parser falls back to the last number (GSM8K) or the last parenthesized option letter (MMLU-Pro), and failed when neither exists, which grades as incorrect. Appendix B shows one

¹Sources: Llama-3.1-8B-Instruct’s generation_config.json, llama.cpp’s defaults, the OpenAI and Anthropic API references (July 2026).

Table 1: The experiments. Both tasks run in every row except Stratified (MMLU-Pro only, fresh items); the BF16 reference run is Qwen3-4B only; the configurations are listed under Decoding. Gen. counts graded generations, or prompts and continuations for the two probes (greedy runs once per item; two Mistral prompts end before the injection point and are skipped).

Experiment	Models	Quantization	Configurations	T	$N \times K$	Gen.	Where
Main grid	7 (5 families)	Q8/Q6/Q4/Q3 (+BF16)	8	0.7, 1.0, 1.3	50×3	185,600	\$4.1, 4.3, 4.4
Higher- T series	Llama-3.1, Qwen2.5, Gemma-3	Q8	4	1.3, 1.5, 1.7, 2.0	50×3	14,400	\$4.4
Additional Llama releases	Llama-3-8B, Llama-3.2-3B	Q8	greedy, temp.	0.7, 1.0, 1.3	50×3	2,000	\$4.1
Engine check	Llama-3.2-3B, Llama-3.1-8B, Qwen3-4B	BF16/INT8 (transformers), F16 (llama.cpp)	greedy, temp.	0.7, 1.0, 1.3	50×3	4,000	\$4.2
Commit bridge	Llama-3.2-3B, Qwen3-4B	Q8 (llama.cpp v0.4.0)	greedy, temp.	0.7, 1.0, 1.3	50×3	2,000	\$4.2
Stratified	Llama-3.1, Qwen2.5, Gemma-3	Q8	greedy, temp.	0.7, 1.0, 1.3	50×3	1,500	\$4.2
High T	Llama-3.1, Gemma-3	Q8	4	2.0, 3.0	15×3	1,440	App. F
Thinking on	Qwen3-8B	Q8	3	0.7, 1.3	25×2	450	App. H
Token-score analysis	7	Q8	greedy	–	50×1	700	App. K
Forced token	7	Q8	greedy, inject.	–	50×1	1,396	App. K

generation for each extraction outcome and checks the rule against markdown formatting. Every comparison in the study compares runs on the same fixed items.

Models and quantization. The grid holds seven instruction-tuned models from five model families: Llama-3.1-8B (Grattafiori et al., 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Qwen2.5-7B (Yang et al., 2024), Qwen3-1.7B/4B/8B (Yang et al., 2025) (thinking off; a thinking-on spot-check in Appendix H), and Gemma-3-12B (Gemma Team, 2025). Each model runs at four quantization levels (Q8_0, Q6_K, Q4_K_M, Q3_K_M), all imatrix GGUF conversions from a single provider so the quantization axis is not confounded by the conversion procedure. Q8_0 is the primary precision for every temperature result in Section 4.1; the sampler comparisons of Section 4.4 pool the four levels, and Appendix I shows the drops level by level.

Replication runs. Two temperature-only runs reuse the grid’s fixed items, chat template, token budgets, and parser, with the greedy baseline and plain temperature at the grid’s three temperatures, $K=3$. The additional Llama evaluation adds two more Llama releases, Llama-3-8B (Meta AI, 2024a) and Llama-3.2-3B (Meta AI, 2024b), at Q8_0 through the same engine. The engine and precision run repeats the axis through a second engine, HF transformers, with plain temperature sampling and no truncation or penalties: Llama-3.2-3B on the official BF16 weights, Llama-3.1-8B in 8-bit integer weights, and Qwen3-4B in BF16 as a robust control; Llama-3.2-3B also runs through llama.cpp on the unquantized GGUF (Section 4.2; settings and provenance in Appendix E).

Statistical inference. All confidence intervals come from a nonparametric bootstrap ($B=2000$; 5000 for the comparisons of Appendix L) clustered on task items. We resample the 50 items instead of individual generations, because the $K=3$ repe-

titions of one item share that item’s difficulty and treating them as independent would understate the uncertainty. For every comparison in the grid, both configurations use the same items, so the bootstrap is paired.

Hardware and reproducibility. The study totals about 213,000 graded generations (Table 1). Hardware and the fixed llama.cpp commit are in Appendix M; model-file hashes are in the repository’s DOWNLOADS.md.

4 Results

4.1 Temperature sensitivity across models

We begin with plain temperature sampling and no truncation, because the accuracy a model loses when its full distribution is sampled at a higher temperature is exactly the failure the truncation samplers of Section 2 exist to prevent. Figure 1 plots the drop from $T=0.7$ to $T=1.3$ at Q8_0 for every model evaluated at both temperatures, and Table 2 gives the values together with the change in two measures of incoherent or incomplete generation: the share of generations that run to the token cap, and the share that end in the specified final-answer sentence (a strict parse, Section 3).

Temperature sensitivity varies continuously across the models. The Qwen3 family and Gemma-3 drop at most 5 points on either task with every CI covering zero; Qwen2.5 and Mistral drop 4 to 9 points, each with one task CI excluding zero; Llama-3.1-8B drops 30 and 38. Its drop begins inside the default temperature range, since from $T=0.7$ to 1.0 its MMLU-Pro accuracy already falls and its run-to-cap rate doubles (Appendix B). Two controls rule out the obvious confounds. A low starting accuracy does not explain the collapse, since Qwen3-1.7B scores 79% on GSM8K, above Llama’s 77%, and does not collapse. Model age does not explain it either, since Mistral-7B-v0.3

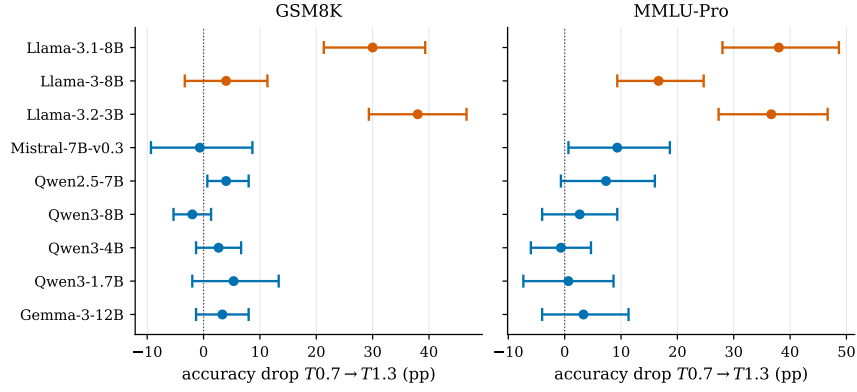


Figure 1: Plain-temperature accuracy drop $T_{0.7} \rightarrow T_{1.3}$ at Q8_0 for the seven grid models and the two additional Llama releases; points with 95% item-clustered bootstrap CIs. Llama-3 releases in orange.

Table 2: Plain-temperature accuracy at $T=0.7$ and 1.3 (Q8_0, $N=50$, $K=3$), the drop with its 95% item-clustered bootstrap CI (bold: lower limit above 15 pp), and the change from $T=0.7$ to 1.3 in the run-to-cap rate and the strict-parse rate.

Model	Task	acc (%) at T		drop [95% CI]		change at 1.3 (pp)	
		0.7	1.3			cap	strict
Llama-3.1-8B	GSM8K	77.3	47.3	+ 30.0	[21.3, 39.3]	+43	-45
	MMLU-Pro	48.0	10.0	+ 38.0	[27.3, 48.7]	+75	-67
Llama-3-8B	GSM8K	75.3	71.3	+4.0	[-2.7, 10.7]	+5	-4
	MMLU-Pro	44.0	27.3	+16.7	[8.7, 24.7]	+29	-32
Llama-3.2-3B	GSM8K	74.0	36.0	+ 38.0	[29.3, 47.3]	+55	-57
	MMLU-Pro	40.0	3.3	+ 36.7	[27.3, 46.7]	+82	-77
Mistral-7B-v0.3	GSM8K	31.3	32.0	-0.7	[-10.0, 8.7]	+1	-1
	MMLU-Pro	29.3	20.0	+9.3	[0.7, 18.7]	+9	-8
Qwen2.5-7B	GSM8K	92.7	88.7	+4.0	[0.7, 8.0]	+1	-4
	MMLU-Pro	65.3	58.0	+7.3	[-1.3, 16.0]	+7	-10
Qwen3-8B	GSM8K	92.7	94.7	-2.0	[-5.3, 1.3]	+1	+2
	MMLU-Pro	60.7	58.0	+2.7	[-4.0, 10.0]	+6	-4
Qwen3-4B	GSM8K	96.0	93.3	+2.7	[-1.3, 7.3]	+0	+0
	MMLU-Pro	64.0	64.7	-0.7	[-6.0, 4.7]	+2	-1
Qwen3-1.7B	GSM8K	79.3	74.0	+5.3	[-2.0, 13.3]	+3	+5
	MMLU-Pro	48.0	47.3	+0.7	[-6.7, 8.0]	+6	-5
Gemma-3-12B	GSM8K	92.0	88.7	+3.3	[-1.3, 8.7]	+2	-2
	MMLU-Pro	63.3	60.0	+3.3	[-4.0, 11.3]	+11	-11

predates Llama-3.1 and is robust.

A single collapsing checkpoint could still be an effect specific to that release, so we repeat the temperature tests on two more Llama releases. Llama-3.2-3B collapses like Llama-3.1-8B (+38.0 pp on GSM8K, +36.7 on MMLU-Pro) and with the same degeneration pattern, with its run-to-cap rate rising by 55 and 82 points. Llama-3-8B lies between the two, with no measurable drop on GSM8K and +16.7 pp on MMLU-Pro, the longer task, a larger drop than any non-Llama model shows. Every model that loses more than 10 points is a Llama-3 release, and the three releases produce the three largest MMLU-Pro drops in the study. What sets them apart from the other four families is a question we discuss in Section 5.

4.2 Replication across item sets and implementations

Every grid result comes from one engine, llama.cpp, on imatrix GGUF quantizations from one provider, so a collapse could in principle be a property of the engine or of the conversion rather than of the model. We therefore repeat the temperature tests on the same fixed item set through a second engine, HF transformers, for both collapsing Llama checkpoints and for one robust control (Table 3). Llama-3.2-3B runs in native BF16 on the official weights and, through llama.cpp, as the unquantized F16 GGUF; Llama-3.1-8B runs in 8-bit integer weights, the highest precision our 16 GB card holds; Qwen3-4B runs in BF16. The collapse replicates on both Llama checkpoints and the control stays flat. In BF16 through transformers Llama-3.2-3B loses 40.0 pp on GSM8K and 36.0 on MMLU-Pro (Q8 through llama.cpp: 38.0 and 36.7), with 61 and 92% of its $T=1.3$ generations at the cap, and the paired difference between the two engines' drops is +2.0 [-9.3, 12.7] on GSM8K and -0.7 [-9.3, 8.0] on MMLU-Pro. Llama-3.1-8B in INT8 loses 43.3 and 42.7 (Q8: 30.0 and 38.0); on GSM8K the INT8 drop exceeds the Q8 drop by 13.3 [2.7, 24.0]. Qwen3-4B in BF16 moves by 1.3 pp and 0.7, as in its GGUF cells. These runs replicate the drop and its failure pattern. They do not show that the engines are equivalent: the intervals on the differences span about 20 pp, and the 8B GSM8K difference excludes zero. For Llama-3.2-3B the prompt strings differ only by a date its template stamps into the system header, which changes no token count; the other two arms reproduce every grid prompt hash (Appendix E).

The 2026 panel (Section 4.1) needs a later

Table 3: Engine, precision, and llama.cpp-release replication on the fixed items, two collapsing Llama checkpoints and a robust control: the plain-temperature drop with its 95% item-clustered bootstrap CI, and the run-to-cap rate at $T=1.3$ on GSM8K / MMLU-Pro. Full accuracy levels in Appendix E.

Model	Engine, precision	drop $T0.7 \rightarrow 1.3$ [95% CI]		cap at 1.3 G / M
		GSM8K	MMLU-Pro	
Llama-3.2-3B	BF16, transformers	+40.0 [30.0, 50.0]	+36.0 [24.7, 47.3]	61 / 92
	F16 GGUF, llama.cpp	+31.3 [22.7, 40.0]	+32.7 [22.7, 42.7]	53 / 91
	Q8 GGUF, llama.cpp v0.4.0	+42.7 [34.0, 52.0]	+34.7 [24.7, 45.4]	63 / 93
	Q8 GGUF, llama.cpp	+38.0 [28.7, 46.7]	+36.7 [28.0, 46.0]	55 / 92
Llama-3.1-8B	INT8, transformers	+43.3 [34.0, 52.7]	+42.7 [32.0, 52.7]	54 / 95
	Q8 GGUF, llama.cpp	+30.0 [22.0, 38.7]	+38.0 [28.0, 48.7]	51 / 86
Qwen3-4B	BF16, transformers	+1.3 [-1.3, 4.0]	-0.7 [-6.0, 5.3]	0 / 13
	Q8 GGUF, llama.cpp v0.4.0	+2.7 [-0.7, 7.3]	+1.3 [-4.7, 8.0]	0 / 11
	Q8 GGUF, llama.cpp	+2.7 [-1.3, 7.3]	-0.7 [-6.0, 4.7]	1 / 12
	BF16 GGUF, llama.cpp	-1.3 [-4.0, 1.3]	-2.0 [-7.3, 3.3]	0 / 15

llama.cpp release than the grid’s pinned build, so we also rerun the Llama-3.2-3B and Qwen3-4B temperature tests on that release (the v0.4.0 rows of Table 3). Both reproduce their pinned-build drops, and each paired difference in drops covers zero (Appendix M gives the release and its settings).

The runs also share the item set. The fixed MMLU-Pro subset is single-category (business), because the test split is sorted by category, so the stratified replication of Appendix C repeats the temperature tests on 50 new items drawn from all 14 categories. It reproduces the ordering, with Llama at +43.3 pp [31.3, 55.3], Qwen2.5 at +6.0, and Gemma at +7.3.

4.3 What fails when accuracy falls

All analyses in this subsection use plain temperature sampling. An accuracy drop can come from two failures: completed answers that are wrong, or generations that never reach an answer. The answer-extraction outcome that Section 3 records for every generation separates the two, so we separate Llama’s accuracy loss by extraction outcome (Figure 2; full decomposition and example outputs in Appendix B).

Nearly all of the drop is the second failure. At $T=1.3$, 53.8% of Llama’s GSM8K generations and 89.2% of its MMLU-Pro generations degenerate into token-level noise and run to the token cap. A blinded audit of 152 sampled outputs, labelled without access to model, temperature, or parser fields, supports this interpretation. Every Llama generation at the cap at $T=1.3$ in the sample is incoherent, its generations at the cap at $T=0.7$ are coherent reasoning cut short by the budget, and the parse failures at $T=0.7$ are formatting failures (Appendix B). Among the generations that do produce a well-formed final answer (a strict parse: the spec-

ified final-answer sentence is present), Llama still scores 97.3% on GSM8K at $T=1.3$. That number needs a caveat. Survival is selective, because easier items produce shorter generations and shorter generations have less room to lose coherence, so the 97.3% is measured on an easier-than-average set of items. The matched comparison is greedy decoding on the same items for which sampling produced a well-formed answer, at the same quantization level, which scores 97.3%, against 78.0% on all items. Among the generations that finish, then, accuracy at $T=1.3$ matches greedy decoding on the same items (97.3 vs. 97.3% on GSM8K; 76.9 vs. 78.8% on MMLU-Pro). The comparison cannot say how the incoherent generations would have scored had they finished. It shows that temperature lowers the rate at which the model finishes and leaves the accuracy of the answers it does finish unchanged.

The two task parsers classify the same failure differently. Degenerate text almost always contains a number, so GSM8K’s last-number fallback grades an incoherent generation as a wrong answer, while MMLU-Pro’s letter parser reports a parse failure. Temperature does not raise Llama’s rate of wrong answers; it raises the rate of generations that never reach one.

4.4 Sampler gains depend on temperature sensitivity

This subsection varies the sampler. For each model, task, and temperature we compare each of the six truncation configurations with plain temperature on the same items, as the paired accuracy difference pooled over the four quantization levels, with an item-clustered bootstrap CI (per-configuration accuracy grids in Appendix A; all comparisons in Appendix L; a generalized estimating equation, a regression that accounts for the repeated observations of each item, fitted over the full grid reproduces every interval to within 0.6 pp). At $T=0.7$ and 1.0, the default temperatures, no configuration gains more than 3.3 pp on any robust model, no pointwise upper 95% limit exceeds 7.5 pp (a simultaneous bound on the largest gain over all 144 comparisons is 7.8), 135 of 144 point estimates lie within 3 pp of plain temperature, and 6 intervals exclude zero (2 above, 4 below), the number chance alone gives. At $T=1.3$, 22 of 72 comparisons lie above zero, all on Mistral, Qwen2.5, and Gemma-3, the three non-Llama models with the largest temperature drops, with gains of 3 to 10 pp, largest on the longer task; no Qwen3 comparison

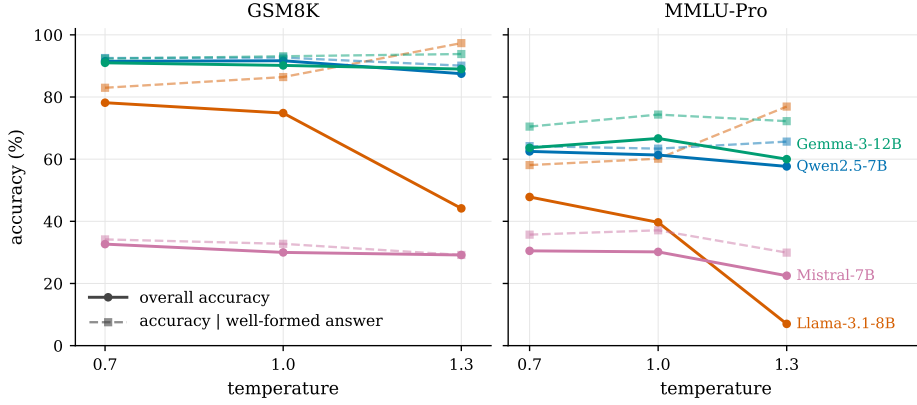


Figure 2: Overall accuracy (solid) versus accuracy among well-formed, strict-parsed answers (dashed) under plain temperature sampling; four models with different levels of temperature sensitivity (the three Qwen3 models maintain stable accuracy and are omitted for legibility). The dashed lines never fall. Among well-formed answers accuracy holds, and Llama’s rises because its completed answers come from easier items (Section 4.3).

leaves zero at any temperature. Restricted to Q8_0 alone (150 generations per cell instead of 600), no point estimate at $T \leq 1.0$ exceeds 4.7 pp and no interval excludes zero, but the upper limits widen to 13 pp, so the pooled analysis gives the smaller upper bound and the Q8 analysis the more conservative one. On the robust models no truncation rule gains more than the bound allows at the default temperatures, and gains of a few points appear once a model has begun to lose accuracy; on Llama the rules that exclude the most tokens already gain 6 to 11 pp on MMLU-Pro at $T=1.0$ (Appendix L), where its collapse has begun.

The order of temperature scaling and truncation changes accuracy by at most 4 pp on the robust models at $T \leq 1.0$ and by up to 7 pp on Mistral at $T=1.3$; on Llama, truncating before temperature scaling preserves more accuracy at $T=1.3$ (top- p 64% $\rightarrow$ 73% on GSM8K), as the effect of that order on the truncation threshold (Section 2) predicts (both orders in Appendix G).

Cliff temperatures. To see whether the robust models also collapse at higher temperature, we test $T \in \{1.3, 1.5, 1.7, 2.0\}$ on three models spanning the gradient of Figure 1: Llama-3.1-8B, Qwen2.5-7B, and Gemma-3-12B (Q8, both tasks, $N=50$, $K=3$). Figure 3 and Table 4 give the accuracy at each temperature and the drops against each model’s own $T=0.7$ baseline. All three reach a cliff inside the tested range, at different temperatures. Taking the cliff as the lowest tested temperature at which a model loses more than half its $T=0.7$ accuracy, the three models collapse on GSM8K in their Figure 1 order: Llama at 1.5,

Qwen2.5 at 1.7, and Gemma at 2.0. At $T=1.5$ the CI of Llama’s drop does not overlap the other two models’, and at $T=1.7$ Qwen2.5’s does not overlap Gemma’s. Within each model the cliff comes at a lower temperature on MMLU-Pro, the task with roughly twice the generation length, as expected if each generated token carries a risk of making the continuation incoherent and that risk accumulates over a longer generation. There Llama has already collapsed at $T=1.3$, and Qwen2.5 and Gemma both collapse at $T=1.5$.

Truncation raises the cliff temperature, and the more tokens a rule excludes the more it raises it: top- p raises the cliff by about 0.2, min- p keeps 70% accuracy at $T=1.5$ on Llama GSM8K where plain temperature is at 9%, and accuracy under top- $n\sigma$ is unchanged up to $T=2.0$ on all three models. The strict-parse rate tracks accuracy almost one-to-one across all tested temperatures, so the sharp fall in accuracy coincides with a sharp fall in well-formed answers. This reconciles the sampler papers with the bound above. Min- p and top- $n\sigma$ are validated at temperatures of 1.5 to 3 (Nguyen et al., 2025; Tang et al., 2024), above every model’s deployment range, where they do improve accuracy (Appendix F).

5 Discussion and conclusion

For practitioners. At deployment temperatures the consequential choice is the model. On a robust one every configuration we test stays within 10 points and most within 3; on a fragile one accuracy varies by 42 points across the same configurations. The fragility can be screened for before deploy-

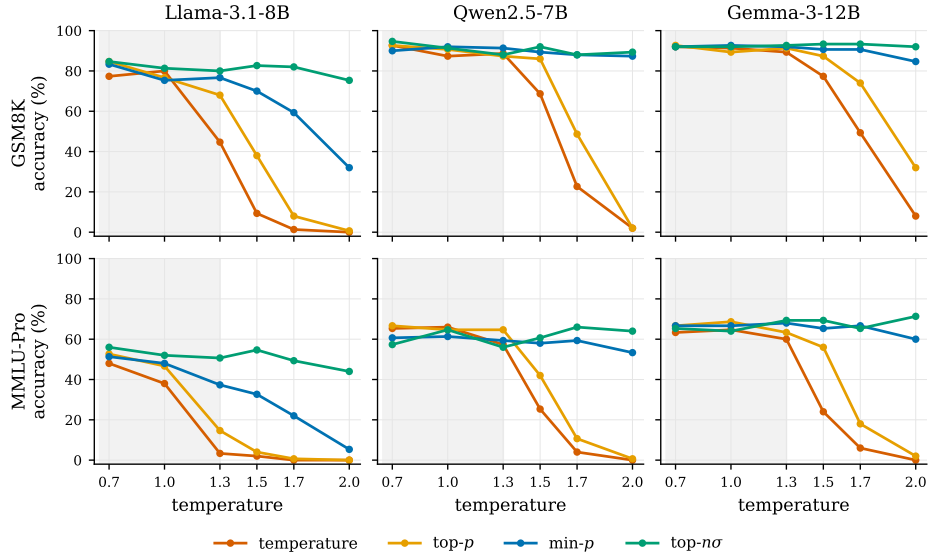


Figure 3: Cliff temperatures. Accuracy versus temperature per sampler (the $T \leq 1.0$ points come from the main grid; all at Q8). Each model has a cliff, at a temperature that differs by model. Truncation raises each model’s cliff temperature.

Table 4: Higher-temperature series: accuracy drop against each model’s own $T=0.7$ baseline under plain temperature (pp, 95% item-clustered bootstrap CIs; Q8). The series is a separate run from the main grid; its $T=1.3$ step differs from Table 2 by amounts within the CIs (Appendix M).

T	Llama-3.1-8B	Qwen2.5-7B	Gemma-3-12B
<i>GSM8K</i>			
1.3	+32.7 [24.7, 41.3]	+4.0 [−0.7, 9.3]	+2.7 [−2.0, 8.0]
1.5	+68.0 [58.0, 76.7]	+24.0 [15.3, 33.3]	+14.7 [8.7, 21.3]
1.7	+76.0 [66.7, 84.7]	+70.0 [62.0, 77.3]	+42.7 [34.7, 50.7]
2.0	+77.3 [67.3, 86.0]	+90.7 [82.7, 97.3]	+84.0 [76.7, 90.7]
<i>MMLU-Pro</i>			
1.3	+44.7 [34.0, 56.0]	+8.0 [0.0, 16.7]	+3.3 [−3.3, 10.0]
1.5	+46.0 [35.3, 57.3]	+40.0 [28.7, 52.0]	+39.3 [29.3, 48.7]
1.7	+48.0 [36.7, 60.0]	+61.3 [48.7, 74.0]	+57.3 [45.3, 68.0]
2.0	+48.0 [36.7, 59.3]	+65.3 [53.3, 78.0]	+63.3 [51.3, 75.3]

ment. On the fifty fixed items, the drop in strict-parse rate and the rise in run-to-cap rate between $T=0.7$ and 1.3 separate the fragile models from the robust ones in every run of this study (Table 2); we have not validated the rule on models outside it. In production, an incoherent generation carries no well-formed answer, so a check on the answer format catches it, while the rate of well-formed but wrong answers changes little.

For the sampler literature. A sampler evaluation should say where its models’ cliffs sit, since a gain measured past one model’s cliff need not exist on a model whose cliff lies further out (Section 4.4).

Toward a mechanism. Two exploratory tests (Appendix K) ask what a fragile model does differently before coherence is lost. Greedy-path entropy and the per-token chance of drawing a token from beyond the model’s own top 20 both rank the seven grid models’ drops at Spearman $\rho = +0.93$, yet neither separates Llama-3.1 from Mistral, which is robust. A forced tail token does. Inserting the token ranked twentieth by probability into text generated with greedy decoding multiplies Llama’s probability of subsequently sampling a low-probability token 2.3-fold, while no robust model’s rises by more than +0.0015 per position. With seven models and one injection configuration these probes check the loss-of-coherence account for consistency and do not test it causally, and what sets a model’s cliff remains open.

Across nine models from five families, the choice of truncation sampler matters far less at deployment temperatures than the model’s own temperature sensitivity. Robust models lose at most a few points under every sampler we test, while two Llama generations lose up to 38 points under plain temperature and a third loses 17 on the longer task, in two engines and at every precision we tried. Each model we test above the deployment range collapses at its own temperature, Llama already inside that range; truncation raises that temperature but changes little below it. These results argue for screening temperature fragility before deployment rather than tuning the sampler after it.

Limitations

Our sample is 50 items per task with $K=3$ repetitions per cell; the small item count is what makes an experiment of 185,600 generations over every combination of factors feasible. Item-clustered bootstrap CIs accompany every main estimate, and the main drops are 2 to 4 times their CI widths. Mistral-7B’s GSM8K accuracy is low (31% at $T=0.7$), which limits the power to detect a further drop; its MMLU-Pro drop is measured well above zero accuracy and its CI excludes zero. The fixed MMLU-Pro subset is single-category (business) because the test split is sorted by category; the stratified $N=50$ replication of Appendix C reproduces the ordering on fresh items, and it is a second sample of 50 items, so it does not estimate the benchmark-wide effect. $T=1.3$ lies above the default temperatures that models and engines set. It is a test temperature that bounds the deployment range from above, and we do not expect deployments to use it.

The scope is open-weight instruction-tuned models of 1.7B to 12B parameters. Whether larger or proprietary models share the Llama family’s fragility is untested. Both tasks are English, zero-shot, and objective, with a single exactly checkable answer, which is what lets the parser measure de-generation. The original use case of truncation samplers, open-ended generation with no checkable answer, lies outside this design, so our bound on sampler gains does not extend to it. We replicate the results across engines and precisions for two Llama checkpoints and one control only, and the Llama-3.1-8B replication runs in 8-bit rather than native precision. We compare samplers only in llama.cpp, at one setting per rule, so the bound applies to those settings and that engine. We have not repeated the sampler comparisons on the stratified subset. Quantization covers the tested llama.cpp formats only, all produced with the same conversion procedure. Thinking mode is off throughout the main grid; the thinking-on spot-check of Appendix H agrees on GSM8K but shows a 10pp MMLU-Pro drop [2, 20] at $N=25$, so our bound does not cover thinking mode.

Every fragile model we observe is a Llama-3 release. The evaluation of additional Llama releases shows the fragility spans three generations, which rules out an effect specific to one checkpoint, but the three share training data and post-training methods, so they are not three independent pieces of evidence. With one fragile family we cannot

tell whether the fragility comes from a training or post-training choice made by Llama’s developers or from something else these models have in common, and we cannot say how common such families are.

Three analysis caveats. Accuracy conditional on a strict parse is subject to survivorship; Section 4.3 bounds it with a matched greedy comparison and does not remove it, and it says nothing about how the incoherent generations would have scored. The bound of Section 4.4 is a pointwise 95% upper limit on the sampler effect at $N=50$ (7.8 pp simultaneous). It does not show that the effect is zero, and it does not establish equivalence within a smaller margin. The mechanism probes correlate seven models and use one injection configuration (a rank-20 token at position 48, a 32-position window); Appendix K presents them as consistency checks, not a causal account, and the grid contains no base-versus-instruct or checkpoint axis, so it cannot say what moves the cliff.

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Table 5: GSM8K accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	78.0	–	–
	temperature	78.2	74.8	44.2
	top- k	79.8	75.2	58.7
	top- p	81.5	78.0	63.8
	min- p	81.2	78.0	72.0
	top- $n\sigma$	78.8	80.7	80.8
	top- p (temp-last)	80.8	78.7	73.0
	min- p (temp-last)	78.7	81.3	76.2
Mistral-7B	greedy ($T=0$)	31.0	–	–
	temperature	32.7	30.0	29.2
	top- k	31.8	31.7	29.8
	top- p	28.2	31.0	34.2
	min- p	32.3	28.3	28.8
	top- $n\sigma$	29.5	30.7	30.2
	top- p (temp-last)	29.5	31.2	29.5
	min- p (temp-last)	29.5	32.0	30.5
Qwen2.5-7B	greedy ($T=0$)	93.5	–	–
	temperature	91.5	91.7	87.5
	top- k	90.2	88.5	88.7
	top- p	92.2	91.2	90.3
	min- p	90.8	90.5	91.0
	top- $n\sigma$	92.7	91.2	90.7
	top- p (temp-last)	90.8	91.3	90.3
	min- p (temp-last)	92.0	90.7	91.3
Gemma-3-12B	greedy ($T=0$)	92.0	–	–
	temperature	91.0	90.2	89.0
	top- k	91.8	91.7	90.7
	top- p	91.5	90.3	90.7
	min- p	91.5	91.3	92.3
	top- $n\sigma$	91.3	91.5	91.0
	top- p (temp-last)	90.2	91.8	91.5
	min- p (temp-last)	91.2	90.8	90.3
Qwen3-1.7B	greedy ($T=0$)	75.5	–	–
	temperature	73.3	73.7	73.7
	top- k	72.8	72.7	73.3
	top- p	73.8	75.0	72.7
	min- p	72.2	71.7	74.0
	top- $n\sigma$	74.3	73.5	72.3
	top- p (temp-last)	73.0	72.0	71.3
	min- p (temp-last)	74.7	73.3	74.5
Qwen3-4B	greedy ($T=0$)	95.2	–	–
	temperature	95.2	94.0	93.1
	top- k	94.0	94.5	94.1
	top- p	95.5	95.5	93.5
	min- p	95.1	94.7	92.9
	top- $n\sigma$	94.8	93.9	93.9
	top- p (temp-last)	94.8	95.2	94.3
	min- p (temp-last)	94.5	93.9	93.9
Qwen3-8B	greedy ($T=0$)	94.5	–	–
	temperature	95.0	94.2	94.5
	top- k	94.8	94.8	93.5
	top- p	94.0	94.3	93.7
	min- p	94.8	95.3	94.7
	top- $n\sigma$	96.3	94.2	94.3
	top- p (temp-last)	93.8	94.0	94.0
	min- p (temp-last)	96.2	94.7	95.3

Table 6: MMLU-Pro accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	54.0	–	–
	temperature	47.8	39.7	7.0
	top- k	51.2	43.3	26.8
	top- p	51.3	46.0	16.0
	min- p	51.2	50.8	39.5
	top- $n\sigma$	51.0	50.5	48.8
	top- p (temp-last)	50.3	46.7	37.5
	min- p (temp-last)	51.2	50.2	43.5
Mistral-7B	greedy ($T=0$)	35.5	–	–
	temperature	30.5	30.2	22.5
	top- k	32.5	32.3	28.3
	top- p	30.2	29.5	25.3
	min- p	33.8	30.7	31.0
	top- $n\sigma$	32.3	31.5	31.5
	top- p (temp-last)	29.2	30.2	32.5
	min- p (temp-last)	33.2	33.3	28.7
Qwen2.5-7B	greedy ($T=0$)	64.0	–	–
	temperature	62.5	61.3	57.7
	top- k	62.5	61.7	57.0
	top- p	65.5	62.8	62.7
	min- p	61.3	63.5	59.7
	top- $n\sigma$	62.0	63.5	61.0
	top- p (temp-last)	63.7	62.7	61.3
	min- p (temp-last)	64.2	63.0	62.3
Gemma-3-12B	greedy ($T=0$)	61.0	–	–
	temperature	63.7	66.7	60.0
	top- k	64.3	62.8	63.7
	top- p	65.7	66.7	63.8
	min- p	65.3	67.0	65.5
	top- $n\sigma$	65.2	65.8	65.8
	top- p (temp-last)	65.8	65.0	65.2
	min- p (temp-last)	65.3	63.7	63.8
Qwen3-1.7B	greedy ($T=0$)	45.5	–	–
	temperature	46.0	45.7	45.3
	top- k	47.5	46.5	45.0
	top- p	47.2	47.0	47.2
	min- p	47.0	47.7	46.2
	top- $n\sigma$	46.8	47.2	46.2
	top- p (temp-last)	47.7	46.8	43.5
	min- p (temp-last)	45.7	46.2	49.0
Qwen3-4B	greedy ($T=0$)	61.6	–	–
	temperature	61.9	62.5	61.2
	top- k	63.5	61.1	61.2
	top- p	62.8	61.3	62.8
	min- p	60.9	62.7	62.9
	top- $n\sigma$	62.4	62.4	61.5
	top- p (temp-last)	62.9	62.7	61.7
	min- p (temp-last)	61.3	63.6	61.5
Qwen3-8B	greedy ($T=0$)	60.5	–	–
	temperature	60.5	59.0	59.5
	top- k	60.2	61.3	57.0
	top- p	60.5	61.3	60.5
	min- p	61.0	59.2	59.7
	top- $n\sigma$	60.0	58.8	58.7
	top- p (temp-last)	60.7	59.5	60.5
	min- p (temp-last)	61.3	60.7	59.3

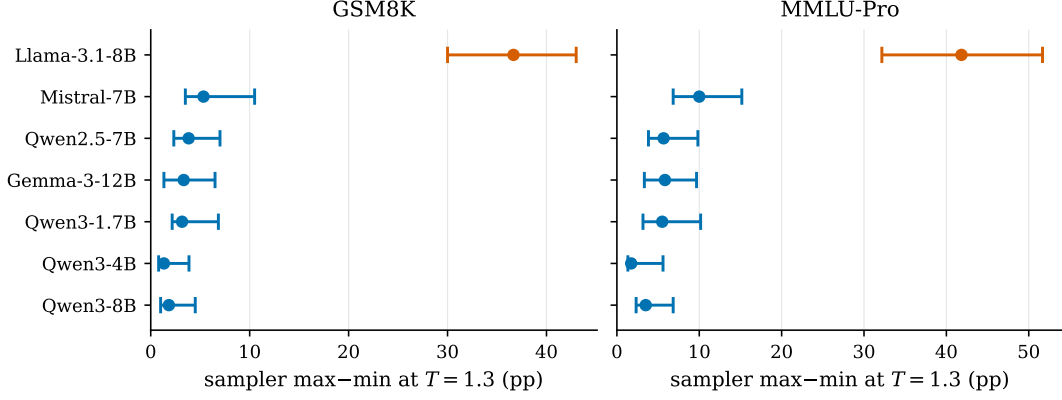


Figure 4: Sampler max–min accuracy spread at $T=1.3$ per model (95% CIs).

[strict] . . . $x = 19.50 / 0.75$ $x = 26$ So
the original price of the book was \$26. The
final answer is \$26.

[flexible] . . . $80 \text{ \texttt{\text{ miles}}} + 150 \text{ \texttt{\text{ miles}}} = 230 \text{ \texttt{\text{ miles}}}$ ### Final Answer:
The final answer is 230.

[failed] . . . $\frac{3,462.20}{35} = 98.92$
Final Answer: The answer is J.

Figure 5: One generation per parse path (endings only). Strict: Llama-3.1-8B, GSM8K, $T=1.3$, graded 26, correct. Flexible: Qwen3-8B, GSM8K, $T=0.7$; the emphasis markers defeat the strict rule and the last-number fall-back grades 230, correct. Failed: Qwen3-8B, MMLU-Pro, $T=0.7$; no parenthesized letter anywhere, graded incorrect (the gold answer is E, so the incorrect grade is correct by coincidence).

[opens] To solve this problem, we’ll use the
gross profit method of inventory evaluation.
Step 1: Calculate the gross profit earned
during the year. . .
[ends] H sieve Con Palette sediment conduit
France called coaching OMAX may PHYS today
Enhanced vertices XOR/t and registration
ordinances efforts yesterday Michael

[opens] To determine the correct type of
research methods, we need to understand the
characteristics of each option: A.
Non-probability: This method involves
selecting. . .
[ends] entageeba lever modeling ign measures
inherits cones quantum auditingippines image
later campaigns Taylorart parte diesel El N
resembled Dental professionals

Figure 6: Two representative degenerate outputs (Llama-3.1-8B, Q8, plain temperature, $T=1.3$, MMLU-Pro; both ran to the 1024-token cap).

in markdown emphasis, which the strict rule does not match; both grade correctly through the flexible rule. A failed parse is a generation with no in-range letter or number at all. On MMLU-Pro that is almost always a incoherent generation, plus 13 to 16% of Mistral’s generations at every temperature that name the option in words rather than by letter, and the markdown case of the third example. Because markdown emphasis defeats the strict rule, we re-graded every record of the main grid with a strict rule that tolerates it: 598 of 185,600 grades change, 556 of them on Qwen3-8B and Qwen3-4B MMLU-Pro; no plain-temperature drop moves by more than 1.0 pp and no deployment-band sampler spread by more than 2.0 pp. The original parser, fixed before any run and mirroring lm-eval-harness, stays primary.

The pattern is the same in each. A coherent opening, a mid-generation loss of coherence, then token noise until the cap. The model never reaches

an answer at all.

Table 7 gives the length-censoring evidence used in Appendix K: run-to-cap rates and the 90th percentile of completion length among *terminated* generations at $T=0.7$ versus $T=1.3$. Robust models are unchanged (ratio 0.85 to 1.19); Llama’s surviving generations shorten ($\times 0.73$ GSM8K, $\times 0.55$ MMLU-Pro), so a incoherent generation does not recover, and a generation survives only if it finishes before its first bad draw.

Blinded outcome audit. To check that the parser’s paths mean what the decomposition takes them to mean, we drew 152 plain-temperature generations at Q8_0 from Llama-3.1-8B, Mistral-7B, Qwen2.5-7B, and Qwen3-8B at $T \in \{0.7, 1.3\}$ on both tasks, stratified by parser path with the uncertain paths oversampled, and had them labelled by a language-model annotator (Claude) that saw

Table 7: Terminated-length censoring under plain temperature at $T=0.7$ versus $T=1.3$ (pooled over quantization). “At cap” includes parse-error-recovered outputs (Appendix B).

Model	Task	at cap (%)		p90 length		ratio
		0.7	1.3	0.7	1.3	
Llama-3.1-8B	GSM8K	5.2	53.8	318	232	$0.73\times$
	MMLU-Pro	11.0	89.2	633	345	$0.55\times$
Mistral-7B	GSM8K	3.5	8.3	367	378	$1.03\times$
	MMLU-Pro	1.0	12.5	628	750	$1.19\times$
Qwen2.5-7B	GSM8K	0.2	1.7	341	361	$1.06\times$
	MMLU-Pro	2.5	10.2	771	732	$0.95\times$
Gemma-3-12B	GSM8K	2.0	5.3	363	344	$0.95\times$
	MMLU-Pro	9.8	17.2	753	763	$1.01\times$
Qwen3-1.7B	GSM8K	2.0	2.2	324	339	$1.05\times$
	MMLU-Pro	15.8	16.3	779	806	$1.03\times$
Qwen3-4B	GSM8K	0.9	0.8	353	361	$1.02\times$
	MMLU-Pro	11.9	12.4	778	731	$0.94\times$
Qwen3-8B	GSM8K	2.3	2.3	379	389	$1.03\times$
	MMLU-Pro	12.3	16.0	850	724	$0.85\times$

only the task and the raw output: completed (a final answer in coherent text), cut off while coherent, degenerate (token incoherent text, script drift, or looping fragments), misformatted (a clear answer the specified sentence would miss), or other. Of the 51 outputs at the token cap, all 8 of Llama’s at $T=1.3$ are degenerate and all 8 at $T=0.7$ are coherent reasoning cut short; the robust models’ 21 cap hits at $T=1.3$ split 8 degenerate, 9 cut off, and 4 with an answer. Of the 26 parse failures, 15 contain a clear answer in an unexpected form (“The answer is **H**.” without parentheses, or a dollar value instead of a letter), 6 are degenerate, and 4 are refusals or “none of the options”; the markdown-tolerant regrade above already covers the first group. Of the 75 strict and flexible parses, 72 are completed or decorated answers, and the annotator marked 34 of the 152 labels as borderline. The rise in the run-to-cap rate from $T=0.7$ to 1.3 is therefore a degeneration signal on Llama, while a raw cap rate at $T=0.7$ is budget censoring, and a parse failure at $T=0.7$ is mostly a formatting failure. Sample, labels, and key are in the repository (results/outcome_audit).

C Stratified MMLU-Pro replication

The fixed $N=50$ MMLU-Pro head subset is single-category (business) because the test split is sorted by category. Table 9 repeats the temperature tests (Q8, $K=3$) on a deterministic stratified subset with

3–4 items from each of the 14 categories. The collapse ordering reproduces.

D Additional Llama releases

This run repeats the temperature tests (greedy baseline plus plain temperature at $T \in \{0.7, 1.0, 1.3\}$) on Llama-3-8B and Llama-3.2-3B: Q8_0, both fixed $N=50$ tasks, $K=3$, 2,000 generations, same pipeline, prompts, and seed derivation as the main grid. Table 10 reports both models next to the Q8-only Llama-3.1-8B slice of the main grid, the same slice Table 2 reports. Greedy baselines are ordinary (Llama-3-8B 76.0% GSM8K / 40.0% MMLU-Pro; Llama-3.2-3B 74.0 / 42.0%). Llama-3.2-3B reproduces the full degeneration pattern at $T=1.3$: strict-parse rate $98.7 \rightarrow 41.3\%$ on GSM8K and $82.7 \rightarrow 6.0\%$ on MMLU-Pro, 55 and 92% of generations at the token cap (counting the streams the server returned with a zero token count, which are degenerate text of cap length), and accuracy conditional on a strict parse at or above the greedy level (85.5 and 55.6%). Llama-3-8B shows the same pattern at a lower rate on MMLU-Pro, with strict-parse $97.3 \rightarrow 65.3\%$ and 29% of generations at the cap, almost all of them the zero-count degenerate streams.

E Engine and precision replication

The transformers configuration loads the official meta-llama/Llama-3.2-3B-Instruct weights (revision 0cb88a4f) in BF16 with transformers 5.16.1 and torch 2.14, samples with do_sample at temperature T with top- k and top- p disabled, repetition penalty 1.0 and no other processor, in batches of eight with left padding; greedy decoding is do_sample=False. Generation stops on the model’s end-of-sequence set (`<|end_of_text|>`, `<|eom_id|>`, `<|eot_id|>`) or at the task’s token budget, and a generation that reaches the budget counts as at cap. The prompt is rendered client-side from the same chat template as the grid. That template stamps the run date into the Llama system header (“Today Date: 06 Sep 2026” against “04 Jul 2026” in the July run), which is the only difference between the two runs’ prompt strings. Re-rendering with the July date reproduces every lineage prompt hash, and the prompt token counts agree on every item. The Llama-3.1-8B configuration loads the unsloth/Llama-3.1-8B-Instruct mirror of the official weights (revision 4699cc75) in 8-bit through bitsandbytes with the same gener-

Table 8: Decomposition of the temperature effect (plain temperature, pooled over quantization; %).

Model	Task	T	acc	strict	failed	at cap	acc strict
Llama-3.1-8B	GSM8K	0.7	78.2	93.0	0.0	5.2	83.0
	GSM8K	1.0	74.8	84.7	0.0	7.8	86.4
	GSM8K	1.3	44.2	43.8	0.5	53.8	97.3
	MMLU-Pro	0.7	47.8	82.3	15.3	11.0	58.1
	MMLU-Pro	1.0	39.7	64.8	29.0	23.7	60.2
	MMLU-Pro	1.3	7.0	8.7	84.5	89.2	76.9
Mistral-7B	GSM8K	0.7	32.7	72.2	0.3	3.5	34.2
	GSM8K	1.0	30.0	67.7	0.0	4.3	32.8
	GSM8K	1.3	29.2	62.8	0.0	8.3	29.2
	MMLU-Pro	0.7	30.5	84.0	12.8	1.0	35.7
	MMLU-Pro	1.0	30.2	80.8	17.0	1.0	37.1
	MMLU-Pro	1.3	22.5	73.5	21.2	12.5	29.9
Qwen2.5-7B	GSM8K	0.7	91.5	98.3	0.0	0.2	92.5
	GSM8K	1.0	91.7	95.7	0.0	0.5	92.7
	GSM8K	1.3	87.5	92.5	0.0	1.7	90.1
	MMLU-Pro	0.7	62.5	97.3	2.3	2.5	64.2
	MMLU-Pro	1.0	61.3	96.5	2.5	3.0	63.4
	MMLU-Pro	1.3	57.7	87.3	10.5	10.2	65.6
Gemma-3-12B	GSM8K	0.7	91.0	98.0	0.0	2.0	92.3
	GSM8K	1.0	90.2	96.8	0.0	3.3	93.1
	GSM8K	1.3	89.0	94.5	0.0	5.3	93.8
	MMLU-Pro	0.7	63.7	90.3	9.0	9.8	70.5
	MMLU-Pro	1.0	66.7	89.7	8.2	10.3	74.3
	MMLU-Pro	1.3	60.0	82.8	14.5	17.2	72.2
Qwen3-1.7B	GSM8K	0.7	73.3	20.3	0.0	2.0	73.8
	GSM8K	1.0	73.7	20.0	0.0	0.7	71.7
	GSM8K	1.3	73.7	19.0	0.0	2.2	71.1
	MMLU-Pro	0.7	46.0	83.7	13.3	15.8	54.8
	MMLU-Pro	1.0	45.7	82.2	15.5	16.3	55.4
	MMLU-Pro	1.3	45.3	83.2	13.5	16.3	54.5
Qwen3-4B	GSM8K	0.7	95.2	94.3	0.0	0.9	95.3
	GSM8K	1.0	94.0	92.3	0.0	0.8	94.8
	GSM8K	1.3	93.1	95.6	0.0	0.8	93.7
	MMLU-Pro	0.7	61.9	87.2	12.5	11.9	70.9
	MMLU-Pro	1.0	62.5	87.3	12.0	10.8	71.6
	MMLU-Pro	1.3	61.2	87.5	11.1	12.4	70.0
Qwen3-8B	GSM8K	0.7	95.0	48.2	0.0	2.3	98.3
	GSM8K	1.0	94.2	49.2	0.0	2.3	98.6
	GSM8K	1.3	94.5	52.2	0.0	2.3	98.1
	MMLU-Pro	0.7	60.5	82.5	15.2	12.3	72.3
	MMLU-Pro	1.0	59.0	80.5	15.5	14.8	72.3
	MMLU-Pro	1.3	59.5	79.8	16.3	16.0	73.5

Table 9: Stratified versus head-of-split (business) MMLU-Pro subset; plain temperature accuracy (%) at Q8_0.

Model	Subset	T			
		0.7	1.0	1.3	drop
Llama-3.1-8B	stratified	44.7	38.7	1.3	+43.3 pp
Llama-3.1-8B	head-50	48.0	38.0	10.0	+38.0 pp
Qwen2.5-7B	stratified	54.7	53.3	48.7	+6.0 pp
Qwen2.5-7B	head-50	65.3	66.0	58.0	+7.3 pp
Gemma-3-12B	stratified	64.0	62.7	56.7	+7.3 pp
Gemma-3-12B	head-50	63.3	64.7	60.0	+3.3 pp

ation settings, in batches of four, and stops on the same end-of-sequence set. The Qwen3-4B configuration loads Qwen/Qwen3-4B (revision 1cfa9a72) in BF16 with thinking disabled in the template, as in the grid, in batches of eight, and stops on `<|im_end|>` or `<|endoftext|>`. Both reproduce every grid prompt hash. The F16 GGUF configuration uses the grid’s fixed llama.cpp build and settings (Appendix M). llama.cpp reports zero completion tokens for a stream it recovered from a parse error on degenerate output; those count as cap hits here, as in the rest of the paper. Table 11 gives accuracy levels, drops, cap rates, and the paired

Table 10: Additional Llama releases: plain-temperature accuracy and $T_{0.7} \rightarrow 1.3$ drop (pp, 95% item-clustered bootstrap CIs), all at Q8_0. The Llama-3.1-8B rows are the Q8-only slice of the main grid.

Model	Task	acc (%) at T		drop [95% CI]
		0.7	1.3	
Llama-3-8B	GSM8K	75.3	71.3	+4.0 [−2.7, 11.3]
	MMLU-Pro	44.0	27.3	+16.7 [9.3, 24.0]
Llama-3.1-8B	GSM8K	77.3	47.3	+30.0 [21.3, 38.7]
	MMLU-Pro	48.0	10.0	+38.0 [28.0, 48.7]
Llama-3.2-3B	GSM8K	74.0	36.0	+38.0 [29.3, 47.3]
	MMLU-Pro	40.0	3.3	+36.7 [26.7, 46.0]

difference of each configuration’s drop from the Q8 llama.cpp cell.

F High-temperature regime

Table 12 reports a small probe ($N=15$, $K=3$, Q8) in the regime where min- p and top- $n\sigma$ report their largest gains. The sampler ordering is top- $n\sigma \gg \text{min-}p \gg \text{top-}p \approx \text{temperature}$; every generation in the zero cells ran to the token cap, and even the most robust model collapses under plain temperature.

G Temperature-first versus temperature-last

Table 13 gives both chain orders for the two probability-space truncators. The two orders are indistinguishable on the robust models. On Llama at $T=1.3$, truncating before temperature scaling is more protective, worth +9.2 pp on GSM8K and +21.5 pp on MMLU-Pro for top- p (+4.2 and +4.0 for min- p); the results section explains why.

H Thinking-mode spot-check

The main grid runs Qwen3 with thinking off. A confirmatory spot-check (Qwen3-8B, Q8, thinking on, 450 generations, $N=25$, $K=2$) reproduces the GSM8K result (94% at $T=0.7$ versus 96% at $T=1.3$ under plain temperature) but not the MMLU-Pro one: 56% versus 46%, a 10 pp drop with item-clustered CI [2, 20] at $N=25$, with min- p flat on both tasks. With thinking on, the longer traces make Qwen3-8B less robust on the longer task, so the thinking-off null of the main grid should not be read as covering thinking mode.

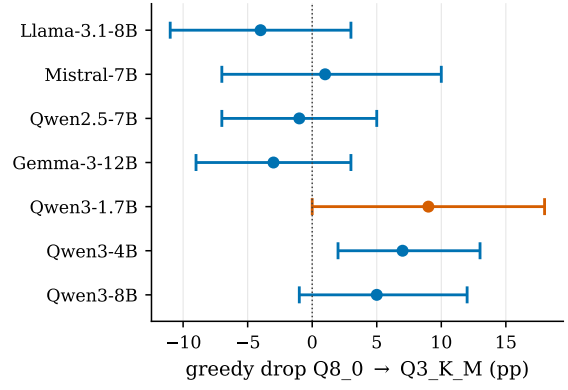


Figure 7: Greedy accuracy drop Q8_0 → Q3_K_M per model (95% CIs), pooled tasks.

I Quantization

The results so far pool over the four quantization levels. For that to be safe, quantization must not cost accuracy on its own, and it must not change what temperature and the samplers do. Quantization costs little accuracy. From Q8_0 down to Q3_K_M, greedy accuracy is unchanged on every model of 7B parameters or more, and a real cost appears only on the two smallest models, 7 pp on Qwen3-4B and 9 pp on Qwen3-1.7B. It changes nothing about the collapse either. Recomputed separately at each level, the plain-temperature drop of Section 4.1 shows Llama-3.1 collapsing at every level, with drops of 30 to 47 pp across levels and tasks, while no robust model exceeds 12 pp at any single level. Llama’s sampler ordering at $T=1.3$ is identical at all four levels (Table 14). Figure 7 plots the direct cost of quantization on greedy accuracy. Table 14 recomputes the plain-temperature drop of Table 2 separately at each quantization level. Each cell pools 150 generations ($N=50$, $K=3$), so single-level estimates are noisier than the pooled ones. The BF16 reference run on Qwen3-4B shows the same null drop as its quantized versions (−1.3 pp GSM8K, −2.0 pp MMLU-Pro), and so does its BF16 run through transformers (+1.3, −0.7; Appendix E).

J Answer agreement across repetitions

A second check separates robustness from determinism. A model with very peaked next-token distributions would also hold its accuracy, because sampling would return the greedy path every time and the three repetitions of an item would come out as near copies. The repetitions show the opposite. No item’s three repetitions are identical

Table 11: Engine and precision replication: plain-temperature accuracy at $T=0.7$ and 1.3, the drop (pp, 95% item-clustered bootstrap CIs), the run-to-cap rate at $T=1.3$, and the difference of the drop from the Q8 llama.cpp cell on the same items.

Model	Engine, precision	Task	acc (%) at T		drop [95% CI]	cap at 1.3	drop – drop(Q8)
			0.7	1.3			
Llama-3.2-3B	BF16, transformers	GSM8K	74.7	34.7	+40.0 [29.3, 49.3]	61	+2.0 [−9.3, 12.7]
	BF16, transformers	MMLU-Pro	40.7	4.7	+36.0 [25.3, 48.0]	92	−0.7 [−9.3, 8.0]
	F16 GGUF, llama.cpp	GSM8K	74.0	42.7	+31.3 [23.3, 40.0]	53	−6.7 [−17.3, 4.0]
	F16 GGUF, llama.cpp	MMLU-Pro	38.0	5.3	+32.7 [22.7, 43.3]	91	−4.0 [−11.3, 2.7]
	Q8 GGUF, llama.cpp v0.4.0	GSM8K	77.3	34.7	+42.7 [34.0, 50.7]	63	+4.7 [−3.3, 12.7]
	Q8 GGUF, llama.cpp v0.4.0	MMLU-Pro	39.3	4.7	+34.7 [24.7, 46.0]	93	−2.0 [−10.0, 5.3]
	Q8 GGUF, llama.cpp	GSM8K	74.0	36.0	+38.0 [29.3, 47.3]	55	–
	Q8 GGUF, llama.cpp	MMLU-Pro	40.0	3.3	+36.7 [27.3, 46.0]	92	–
Llama-3.1-8B	INT8, transformers	GSM8K	82.7	39.3	+43.3 [34.0, 52.7]	54	+13.3 [3.3, 23.3]
	INT8, transformers	MMLU-Pro	45.3	2.7	+42.7 [32.0, 52.7]	95	+4.7 [−3.3, 12.7]
	Q8 GGUF, llama.cpp	GSM8K	77.3	47.3	+30.0 [22.0, 38.7]	51	–
	Q8 GGUF, llama.cpp	MMLU-Pro	48.0	10.0	+38.0 [28.0, 49.3]	86	–
Qwen3-4B	BF16, transformers	GSM8K	94.7	93.3	+1.3 [−1.3, 4.0]	0	−1.3 [−5.3, 2.0]
	BF16, transformers	MMLU-Pro	60.7	61.3	−0.7 [−6.0, 4.7]	13	−0.0 [−7.3, 6.7]
	Q8 GGUF, llama.cpp v0.4.0	GSM8K	96.7	94.0	+2.7 [−0.7, 7.3]	0	+0.0 [−4.0, 4.0]
	Q8 GGUF, llama.cpp v0.4.0	MMLU-Pro	65.3	64.0	+1.3 [−4.7, 7.3]	11	+2.0 [−6.0, 10.0]
	Q8 GGUF, llama.cpp	GSM8K	96.0	93.3	+2.7 [−1.3, 6.7]	1	–
	Q8 GGUF, llama.cpp	MMLU-Pro	64.0	64.7	−0.7 [−6.0, 4.7]	12	–
	BF16 GGUF, llama.cpp	GSM8K	95.3	96.7	−1.3 [−4.0, 1.3]	0	−4.0 [−10.0, 2.0]
	BF16 GGUF, llama.cpp	MMLU-Pro	58.0	60.0	−2.0 [−7.3, 3.3]	15	−1.3 [−8.0, 6.0]

Table 12: Accuracy (%) at $T=2$ and $T=3$ (45 observations per cell).

Task	Sampler	Llama-3.1-8B		Gemma-3-12B	
		$T=2$	$T=3$	$T=2$	$T=3$
GSM8K	temperature	0	0	16	0
	top- p	0	0	27	0
	min- p	33	0	78	24
	top- $n\sigma$	71	80	87	89
MMLU-Pro	temperature	0	0	0	0
	top- p	0	0	0	0
	min- p	0	0	58	0
	top- $n\sigma$	47	51	69	62

on any model, yet the robust models still converge on the answer, 1.1 to 1.4 distinct parsed answers per item on GSM8K at $T=1.3$ (Mistral, whose low GSM8K accuracy scatters its answers, is 2.5). Llama-3.1 rises from 1.5 to 2.3 under the same change, as each repetition becomes incoherent on its own (Table 15). Neither quantization nor determinism explains the pattern. Mean distinct parsed answers per item across the $K=3$ repetitions under plain temperature (Table 15). Robust models sample diverse token sequences that converge on the same answer; a parse failure counts as one distinct answer, so the statistic is a divergence proxy; it is the per-item agreement that self-consistency decoding (Wang et al., 2023) exploits by majority vote. The parse-failure convention censors Llama-3.1 on

MMLU-Pro at $T=1.3$: with 84.5% of generations failing to parse, the repetitions agree on having no answer, and the statistic falls to 1.28. Mistral sits near 2.2 distinct answers at both temperatures, and its scatter has a different source. At a greedy GSM8K accuracy of 31%, most repetitions are wrong, and wrong repetitions rarely agree on the same wrong number at any temperature. The shift attributable to temperature is +0.27.

K Mechanism probes

If the collapse is a loss of coherence process, fragile models should carry more per-token uncertainty before any loss of coherence happens. To measure that uncertainty we decode the 100 fixed prompts (the 50 items of each task) greedily on every model at Q8 and record the raw top-20 next-token probabilities at each position. Each prompt is summarized by four statistics of these distributions: an entropy lower bound (only the top 20 probabilities are observed), the top-1 probability and its margin over the top-2, and the fraction of positions where the top-1 probability falls below 0.5. We decode greedily so that the statistics describe the text each model produces when no random draw is involved, before any random draw can become incoherent it.

Mean greedy-path entropy ranks the seven models’ temperature drops at Spearman $\rho = +0.93$

Table 13: Accuracy (%) with temperature applied before (left) versus after (right) truncation, pooled over quantization.

Model	Method	temperature-first			temperature-last		
		0.7	1.0	1.3	0.7	1.0	1.3
<i>GSM8K</i>							
Llama-3.1-8B	top- <i>p</i>	81.5	78.0	63.8	80.8	78.7	73.0
Llama-3.1-8B	min- <i>p</i>	81.2	78.0	72.0	78.7	81.3	76.2
Mistral-7B	top- <i>p</i>	28.2	31.0	34.2	29.5	31.2	29.5
Mistral-7B	min- <i>p</i>	32.3	28.3	28.8	29.5	32.0	30.5
Qwen2.5-7B	top- <i>p</i>	92.2	91.2	90.3	90.8	91.3	90.3
Qwen2.5-7B	min- <i>p</i>	90.8	90.5	91.0	92.0	90.7	91.3
Gemma-3-12B	top- <i>p</i>	91.5	90.3	90.7	90.2	91.8	91.5
Gemma-3-12B	min- <i>p</i>	91.5	91.3	92.3	91.2	90.8	90.3
Qwen3-1.7B	top- <i>p</i>	73.8	75.0	72.7	73.0	72.0	71.3
Qwen3-1.7B	min- <i>p</i>	72.2	71.7	74.0	74.7	73.3	74.5
Qwen3-4B	top- <i>p</i>	95.5	95.5	93.5	94.8	95.2	94.3
Qwen3-4B	min- <i>p</i>	95.1	94.7	92.9	94.5	93.9	93.9
Qwen3-8B	top- <i>p</i>	94.0	94.3	93.7	93.8	94.0	94.0
Qwen3-8B	min- <i>p</i>	94.8	95.3	94.7	96.2	94.7	95.3
<i>MMLU-Pro</i>							
Llama-3.1-8B	top- <i>p</i>	51.3	46.0	16.0	50.3	46.7	37.5
Llama-3.1-8B	min- <i>p</i>	51.2	50.8	39.5	51.2	50.2	43.5
Mistral-7B	top- <i>p</i>	30.2	29.5	25.3	29.2	30.2	32.5
Mistral-7B	min- <i>p</i>	33.8	30.7	31.0	33.2	33.3	28.7
Qwen2.5-7B	top- <i>p</i>	65.5	62.8	62.7	63.7	62.7	61.3
Qwen2.5-7B	min- <i>p</i>	61.3	63.5	59.7	64.2	63.0	62.3
Gemma-3-12B	top- <i>p</i>	65.7	66.7	63.8	65.8	65.0	65.2
Gemma-3-12B	min- <i>p</i>	65.3	67.0	65.5	65.3	63.7	63.8
Qwen3-1.7B	top- <i>p</i>	47.2	47.0	47.2	47.7	46.8	43.5
Qwen3-1.7B	min- <i>p</i>	47.0	47.7	46.2	45.7	46.2	49.0
Qwen3-4B	top- <i>p</i>	62.8	61.3	62.8	62.9	62.7	61.7
Qwen3-4B	min- <i>p</i>	60.9	62.7	62.9	61.3	63.6	61.5
Qwen3-8B	top- <i>p</i>	60.5	61.3	60.5	60.7	59.5	60.5
Qwen3-8B	min- <i>p</i>	61.0	59.2	59.7	61.3	60.7	59.3

Table 14: Plain-temperature accuracy drop $T_{0.7} \rightarrow 1.3$ (pp) per quantization level.

Model	Q8_0	Q6_K	Q4_K_M	Q3_K_M
<i>GSM8K</i>				
Llama-3.1-8B	+30.0	+32.0	+36.0	+38.0
Mistral-7B	−0.7	−0.7	+3.3	+12.0
Qwen2.5-7B	+4.0	+2.7	+2.7	+6.7
Gemma-3-12B	+3.3	+2.7	+0.0	+2.0
Qwen3-1.7B	+5.3	−6.0	−0.7	+0.0
Qwen3-4B	+2.7	+0.7	+6.0	+2.7
Qwen3-8B	−2.0	+1.3	+2.0	+0.7
<i>MMLU-Pro</i>				
Llama-3.1-8B	+38.0	+46.7	+38.0	+40.7
Mistral-7B	+9.3	+5.3	+10.7	+6.7
Qwen2.5-7B	+7.3	+6.0	+4.0	+2.0
Gemma-3-12B	+3.3	+5.3	+2.7	+3.3
Qwen3-1.7B	+0.7	+6.0	−8.0	+4.0
Qwen3-4B	−0.7	+0.0	+2.0	+4.0
Qwen3-8B	+2.7	−0.7	+0.0	+2.0

Table 15: Mean distinct parsed answers per item ($K=3$, pooled over quantization).

Model	GSM8K		MMLU-Pro	
	$T=0.7$	$T=1.3$	$T=0.7$	$T=1.3$
Llama-3.1-8B	1.51	2.33	1.71	1.28
Mistral-7B	2.21	2.48	1.97	2.21
Qwen2.5-7B	1.10	1.27	1.44	1.57
Gemma-3-12B	1.12	1.16	1.33	1.45
Qwen3-1.7B	1.41	1.43	1.67	1.73
Qwen3-4B	1.05	1.14	1.33	1.38
Qwen3-8B	1.09	1.13	1.41	1.40

(flat-position fraction +0.93; top-1 margin −0.93; Figure 8). With seven models a rank correlation is fragile, so we report an item-clustered bootstrap interval, $[+0.54, +0.96]$, and add an exact permutation test ($p = 0.007$) and leave-one-model-out

ranges. The summaries cannot separate one pair, though. Llama-3.1 and Mistral are nearly tied on every statistic we record, 0.33 versus 0.32 nats of entropy and 1.0×10^{-3} probability mass beyond the top 20 for both, yet Llama drops 40.8 pp on MMLU-Pro and Mistral 8.0. Uncertainty on the greedy path predicts which models drop most, but not by how much.

The probe also estimates the per-token risk that drives a loss of coherence. For each position we compute the probability that sampling at $T=1.3$

picks a token from beyond the model’s own top 20; we call this the escape hazard (the estimator and its tail assumption are stated below). Llama’s is 0.0033 per position and Mistral’s 0.0034, about 0.85 expected tail draws per 256 positions for both, and across the seven models the hazard ranks the drops at the same $\rho = +0.93$ (bootstrap $[+0.57, +0.96]$). Generation lengths show that a incoherent generation does not recover. On every robust model the length distribution of finished generations is unchanged from $T=0.7$ to $T=1.3$ (90th-percentile ratio 0.85 to 1.19), and Llama’s finished generations at $T=1.3$ are *shorter*, because only the short ones finish (Section 4.3). Llama and Mistral therefore draw tail tokens equally often, and a generation that becomes incoherent does not come back, so what separates them must be how often a tail token starts a loss of coherence, that is, what happens after that token is in the context.

A perturbation experiment measures that directly. For each model and fixed prompt we greedily-decode a 48-token prefix, replace the next token with the model’s own rank-20 candidate at that position (median raw probability 2×10^{-9} , a genuine tail draw), and continue greedily for 256 tokens, alongside a control continuation of the unedited prefix. We then compare the two continuations’ escape hazard over the 32 positions after the injection.

Beyond the injected token itself, the continuations show no visible damage. They stay coherent, and Llama’s reach the token cap no more often than its controls (14.0% for both on GSM8K). The difference is in the hazard. Llama’s rises to 2.3 times its baseline ($+0.0042$ per position), Mistral’s to 1.35 times ($+0.0012$), and no robust model gains more than $+0.0015$ (the absolute increase rather than the ratio, for the reason given below). This per-model increase ranks the temperature drops at $\rho = +0.89$ (bootstrap $[+0.39, +0.96]$, permutation $p = 0.012$; the entropy increase, $+0.86$). With seven models these probes check the loss of coherence account for consistency and do not confirm it causally. In short, Llama and Mistral draw tail tokens at the same rate, and what sets Llama apart is how much one tail token raises the chance of the next.

Figure 8 plots greedy-path entropy against the temperature drop. Table 16 consolidates the per-model mechanism evidence of Appendix K: greedy-path entropy lower bound and flat-position fraction ($\text{top-1} < 0.5$) from the summary probe; the

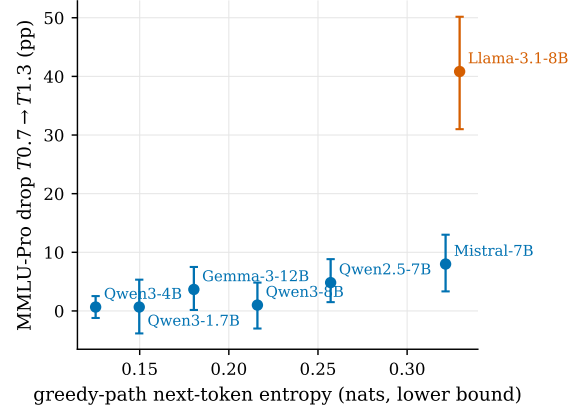


Figure 8: Per-token uncertainty on the model’s own greedy path versus the temperature drop (MMLU-Pro, 95% CIs). The rank correlation is $\rho = +0.93$; the Llama–Mistral gap at equal entropy shows flatness alone does not set the magnitude.

temperature-scaled beyond-top-20 escape hazard per position at $T=1.3$ from the raw top-20 arrays; and the perturbation deltas (post-injection minus control over 32 positions, escape hazard and entropy) from the forced-token experiment. Escape columns are $\times 10^{-3}$. The escape hazard at temperature T is computed from the raw top-20 probabilities $p_1 \geq \dots \geq p_{20}$ and the residual mass $t = 1 - \sum_i p_i$ as $S_{\text{tail}} / (S_{\text{top}} + S_{\text{tail}})$ with $S_{\text{top}} = \sum_i p_i^{1/T}$; the scaled tail mass is not identifiable from the top 20, so we assume the tail consists of t/p_{20} tokens of probability p_{20} , $S_{\text{tail}} = (t/p_{20}) p_{20}^{1/T}$. The one-token lower bound $S_{\text{tail}} = t^{1/T}$ ranks the models identically ($\rho = +0.93$). Leaving out any one model, ρ ranges over $[+0.89, +0.94]$ for entropy and for the hazard, and over $[+0.83, +1.00]$ for the perturbation response. Every static column ties Llama-3.1 with Mistral; only the perturbation response separates them. We use the absolute post-injection hazard increase because survival over L tokens scales as $(1-h)^L$ in the absolute hazard h ; it ranks the drops at $\rho = +0.89$, while the ratio of injected to control hazard does not ($\rho = +0.11$), because Gemma-3 and Qwen3-4B triple a hazard that starts far below Llama’s baseline.

L Bound on the sampler effect

Table 17 gives every paired comparison behind Section 4.4: for each model, task, and temperature, the accuracy of each of the six truncation configurations minus plain temperature on the same

Table 16: Mechanism probes per model (Q8, 100 fixed prompts). Escape hazards are at $T=1.3$; drop is the MMLU-Pro plain-temperature $T_{0.7} \rightarrow 1.3$ accuracy drop (pp), pooled over quantization.

Model	H_{lb}	flat%	esc	Δ_{esc}	ΔH	drop
Llama-3.1-8B	0.329	7.1	3.26	+4.20	+0.110	+40.8
Mistral-7B	0.322	6.8	3.40	+1.19	+0.044	+8.0
Qwen2.5-7B	0.257	4.8	1.97	+1.00	+0.049	+4.8
Gemma-3-12B	0.180	3.0	1.44	+1.42	+0.055	+3.7
Qwen3-1.7B	0.150	2.1	0.36	-0.00	+0.012	+0.7
Qwen3-4B	0.125	1.5	0.22	+0.21	+0.021	+0.7
Qwen3-8B	0.216	3.8	0.71	+0.22	+0.024	+1.0

50 items (each item’s accuracy is its mean over $K=3$ repetitions and the quantization levels, five for Qwen3-4B), with a 95% item-clustered bootstrap CI ($B=5000$). Entries whose interval lies above zero are in bold. A generalized estimating equation over the same records, with an identity link, item clusters, and an exchangeable working correlation, gives the same point estimates and intervals that differ by at most 0.6 pp (median 0.1); scripts `sampler_bound.py` and `sampler_gee.py`.

M Reproducibility

- **Inference stack.** llama.cpp at commit 5aba5364 (server build 9456), native /completion endpoint, CUDA builds of the same commit per GPU architecture (sm_75 for RTX 2080 Ti, sm_120 for RTX 5070 Ti). Server-side BOS insertion is disabled; the client-rendered chat template owns the special tokens, and each record stores the SHA-256 of the exact rendered prompt. The 2026 panel and its bridge run (Section 4.2) use release v0.4.0 (commit 5266f24d, sm_120), which two panel models require, with the same client-side rendering and the server’s template parser off (`-no-jinja`, the pinned build’s default). That release forces its own BOS for Gemma 4, so the Gemma-4 prompt is rendered without one and carries a single BOS, and it reports the cap-length degenerate streams the pinned build returned with a zero token count as an HTTP error without text; the runner records those as zero-token cap hits in both builds, so the only difference is that their text is missing on v0.4.0 runs (their parse path is “failed” rather than “flexible”).

- **Quantizations.** Every quantized checkpoint

is a Bartowski imatrix GGUF (one provider, one recipe, so the quantization axis is not confounded by recipe). SHA-256 hashes of all 31 model files are pinned in the repository (DOWNLOADS.md).

- **Decoding.** Exactly one truncation parameter per request; the sampler chain order is pinned per configuration through the server’s samplers array and recorded in every record. At the pinned commit the server instantiates only the samplers named in that array and then samples from the result, so its own defaults for unlisted samplers (top- k 40, top- p 0.95, min- p 0.05, repetition penalty 1.0) are inert; no penalty, DRY, XTC, or logit-bias parameter is ever sent. Sampler values: top- p 0.95, top- k 40, min- p 0.05, top- $n\sigma$ 1.0 (each method’s recommended value).
- **Seeding.** Every generation’s seed is derived deterministically from (global seed 0, model, quantization, sampler, temperature, task, item, repetition). Greedy decoding is the deterministic $T=0$ baseline with $K=1$.
- **Tasks.** GSM8K (openai/gsm8k, test split, first 50 items) and MMLU-Pro (TIGER-Lab/MMLU-Pro, test split, first 50 items; stratified replication in Appendix C). Parsers mirror lm-eval-harness; parse failures grade as incorrect and are reported.
- **Records.** Each generation is one JSONL record carrying the full identity tuple, the request parameters, the raw output, the parse path, and throughput covariates; runs are resumable by record id. Batched serving (≤ 8 slots) is seed-logged but not bit-exact.
- **Code and data.** The full pipeline, configs, analysis scripts, and every generation record are in an anonymized repository at <https://anonymous.4open.science/r/decoding-robustness-027C>.

Table 17: Paired accuracy difference of each truncation configuration against plain temperature (pp, 95% item-clustered bootstrap CIs; pooled over quantization). Bold: interval above zero.

Model	Task	T	top- p	top- k	min- p	top- $n\sigma$	top- p (T last)	min- p (T last)
Llama-3.1-8B	GSM8K	0.7	+3.3 [+0.2, +6.8]	+1.7 [-1.0, +4.5]	+3.0 [-0.8, +7.0]	+0.7 [-2.7, +4.2]	+2.7 [-0.5, +6.3]	+0.5 [-2.3, +3.2]
		1.0	+3.2 [-0.5, +7.0]	+0.3 [-3.8, +4.3]	+3.2 [-0.7, +7.0]	+5.8 [+2.3, +9.5]	+3.8 [-0.0, +7.8]	+6.5 [+3.0, +10.0]
		1.3	+19.7 [+14.8, +24.5]	+14.5 [+10.0, +19.3]	+27.8 [+22.3, +33.3]	+36.7 [+30.0, +43.3]	+28.8 [+23.3, +34.7]	+32.0 [+25.7, +38.3]
	MMLU-Pro	0.7	+3.5 [+0.3, +6.8]	+3.3 [-0.5, +7.3]	+3.3 [-0.8, +7.7]	+3.2 [-0.2, +6.7]	+2.5 [-1.0, +6.0]	+3.3 [+0.2, +6.7]
		1.0	+6.3 [+2.0, +11.3]	+3.7 [-1.2, +8.5]	+11.2 [+6.7, +16.0]	+10.8 [+6.0, +16.0]	+7.0 [+3.0, +11.2]	+10.5 [+6.2, +14.8]
		1.3	+9.0 [+5.3, +13.3]	+19.8 [+13.5, +26.8]	+32.5 [+24.5, +41.0]	+41.8 [+31.5, +52.3]	+30.5 [+23.0, +38.5]	+36.5 [+27.5, +45.7]
Mistral-7B	GSM8K	0.7	-4.5 [-7.7, -1.3]	-0.8 [-5.7, +3.8]	-0.3 [-3.8, +3.2]	-3.2 [-7.3, +1.2]	-3.2 [-7.5, +1.2]	-3.2 [-6.8, +0.7]
		1.0	+1.0 [-2.7, +4.7]	+1.7 [-1.5, +4.8]	-1.7 [-4.8, +1.7]	+0.7 [-3.0, +4.0]	+1.2 [-3.2, +5.3]	+2.0 [-2.0, +5.8]
		1.3	+5.0 [+0.7, +9.3]	+0.7 [-4.3, +5.5]	-0.3 [-3.8, +3.5]	+1.0 [-2.8, +4.8]	+0.3 [-4.8, +5.0]	+1.3 [-2.3, +5.0]
	MMLU-Pro	0.7	-0.3 [-4.7, +3.8]	+2.0 [-2.3, +6.5]	+3.3 [-0.3, +7.0]	+1.8 [-2.3, +6.2]	-1.3 [-4.7, +2.2]	+2.7 [-1.8, +7.5]
		1.0	-0.7 [-4.7, +3.2]	+2.2 [-1.2, +5.7]	+0.5 [-2.8, +3.8]	+1.3 [-3.2, +5.8]	-0.0 [-4.2, +4.2]	+3.2 [-0.8, +7.3]
		1.3	+2.8 [-1.2, +7.0]	+5.8 [+2.7, +9.2]	+8.5 [+4.5, +12.7]	+9.0 [+4.5, +13.5]	+10.0 [+5.2, +15.2]	+6.2 [+2.2, +10.3]
Qwen2.5-7B	GSM8K	0.7	+0.7 [-1.3, +2.8]	-1.3 [-3.8, +0.8]	-0.7 [-2.0, +0.7]	+1.2 [-0.7, +3.3]	-0.7 [-3.0, +1.5]	+0.5 [-1.2, +2.2]
		1.0	-0.5 [-2.8, +1.7]	-3.2 [-6.0, -0.5]	-1.2 [-3.7, +1.2]	-0.5 [-2.8, +1.7]	-0.3 [-3.3, +2.5]	-1.0 [-3.7, +1.7]
		1.3	+2.8 [+0.7, +5.2]	+1.2 [-0.7, +3.3]	+3.5 [+0.8, +6.3]	+3.2 [+1.2, +5.5]	+2.8 [+0.5, +5.3]	+3.8 [+1.5, +6.5]
	MMLU-Pro	0.7	+3.0 [+0.3, +5.8]	+0.0 [-2.7, +2.7]	-1.2 [-4.5, +2.2]	-0.5 [-3.7, +2.7]	+1.2 [-1.3, +3.5]	+1.7 [-1.2, +4.5]
		1.0	+1.5 [-1.7, +5.0]	+0.3 [-1.8, +2.5]	+2.2 [-1.0, +5.5]	+2.2 [-0.8, +5.3]	+1.3 [-1.0, +3.7]	+1.7 [-1.3, +4.8]
		1.3	+5.0 [+1.7, +8.3]	-0.7 [-3.8, +2.2]	+2.0 [-1.8, +6.2]	+3.3 [-0.3, +6.8]	+3.7 [+0.3, +7.3]	+4.7 [+1.0, +8.7]
Gemma-3-12B	GSM8K	0.7	+0.5 [-1.3, +2.5]	+0.8 [-1.0, +3.2]	+0.5 [-1.3, +2.5]	+0.3 [-1.7, +2.5]	-0.8 [-3.8, +2.0]	+0.2 [-2.2, +2.3]
		1.0	+0.2 [-3.0, +3.2]	+1.5 [-0.5, +3.7]	+1.2 [-1.3, +3.7]	+1.3 [-1.0, +3.7]	+1.7 [-1.0, +4.8]	+0.7 [-1.5, +2.8]
		1.3	+1.7 [-0.2, +3.8]	+1.7 [-0.7, +4.2]	+3.3 [+0.5, +6.7]	+2.0 [-0.0, +4.5]	+2.5 [+0.3, +4.8]	+1.3 [-0.8, +3.5]
	MMLU-Pro	0.7	+2.0 [-0.2, +4.3]	+0.7 [-2.0, +3.3]	+1.7 [-0.8, +4.3]	+1.5 [-1.3, +4.3]	+2.2 [-1.0, +5.5]	+1.7 [-2.2, +5.5]
		1.0	-0.0 [-2.8, +2.8]	-3.8 [-7.2, -0.8]	+0.3 [-2.2, +2.7]	-0.8 [-3.2, +1.3]	-1.7 [-4.8, +1.3]	-3.0 [-6.2, -0.2]
		1.3	+3.8 [+0.3, +8.3]	+3.7 [+0.5, +7.3]	+5.5 [+2.7, +8.7]	+5.8 [+2.2, +10.0]	+5.2 [+2.2, +8.3]	+3.8 [+0.7, +7.3]
Qwen3-8B	GSM8K	0.7	-1.0 [-3.3, +0.8]	-0.2 [-1.5, +1.2]	-0.2 [-1.7, +1.2]	+1.3 [-0.8, +3.5]	-1.2 [-3.5, +0.8]	+1.2 [-0.8, +3.5]
		1.0	+0.2 [-1.7, +2.0]	+0.7 [-1.8, +3.3]	+1.2 [-0.8, +3.3]	-0.0 [-1.8, +1.7]	-0.2 [-2.2, +2.0]	+0.5 [-1.8, +2.8]
		1.3	-0.8 [-2.3, +0.7]	-1.0 [-2.7, +0.5]	+0.2 [-1.8, +2.5]	-0.2 [-1.8, +1.5]	-0.5 [-2.2, +1.0]	+0.8 [-0.3, +2.2]
	MMLU-Pro	0.7	-0.0 [-3.0, +3.2]	-0.3 [-2.7, +2.0]	+0.5 [-2.5, +3.7]	-0.5 [-3.0, +1.8]	+0.2 [-2.7, +2.8]	+0.8 [-2.2, +3.8]
		1.0	+2.3 [-0.7, +5.2]	+2.3 [-0.7, +5.5]	+0.2 [-3.0, +3.5]	-0.2 [-3.0, +2.7]	+0.5 [-2.3, +3.3]	+1.7 [-1.2, +4.7]
		1.3	+1.0 [-1.8, +3.8]	-2.5 [-5.8, +0.7]	+0.2 [-3.2, +3.3]	-0.8 [-4.2, +2.3]	+1.0 [-1.8, +3.8]	-0.2 [-3.3, +3.2]
Qwen3-4B	GSM8K	0.7	+0.3 [-0.9, +1.6]	-1.2 [-2.9, +0.7]	-0.1 [-1.3, +1.3]	-0.4 [-1.6, +0.9]	-0.4 [-2.0, +1.6]	-0.7 [-2.5, +1.6]
		1.0	+1.5 [+0.0, +3.1]	+0.5 [-0.7, +1.7]	+0.7 [-1.1, +2.7]	-0.1 [-1.5, +1.1]	+1.2 [+0.0, +2.4]	-0.1 [-1.5, +1.3]
		1.3	+0.4 [-1.2, +2.0]	+1.1 [-0.7, +2.9]	-0.1 [-1.9, +1.6]	+0.8 [-1.1, +2.7]	+1.2 [-0.5, +3.1]	+0.8 [-0.7, +2.5]
	MMLU-Pro	0.7	+0.9 [-1.6, +3.2]	+1.6 [-0.5, +3.9]	-0.9 [-3.6, +1.5]	+0.5 [-1.6, +2.5]	+1.1 [-0.9, +3.2]	-0.5 [-3.1, +1.9]
		1.0	-1.2 [-3.2, +0.8]	-1.5 [-3.5, +0.4]	+0.1 [-2.7, +3.2]	-0.1 [-2.9, +2.4]	+0.1 [-2.8, +3.2]	+1.1 [-1.5, +3.6]
		1.3	+1.6 [-1.1, +4.5]	+0.0 [-2.8, +2.5]	+1.7 [-0.7, +4.1]	+0.3 [-2.3, +2.7]	+0.5 [-2.1, +3.2]	+0.3 [-2.3, +2.8]
Qwen3-1.7B	GSM8K	0.7	+0.5 [-2.8, +3.8]	-0.5 [-3.8, +2.8]	-1.2 [-4.7, +2.3]	+1.0 [-2.3, +4.3]	-0.3 [-3.0, +2.3]	+1.3 [-1.5, +4.3]
		1.0	+1.3 [-2.0, +4.5]	-1.0 [-4.2, +2.0]	-2.0 [-4.3, +0.2]	-0.2 [-2.7, +2.3]	-1.7 [-4.2, +0.7]	-0.3 [-3.2, +2.5]
		1.3	-1.0 [-4.5, +2.2]	-0.3 [-3.3, +2.8]	+0.3 [-3.0, +3.7]	-1.3 [-4.2, +1.2]	-2.3 [-6.0, +1.2]	+0.8 [-2.2, +3.7]
	MMLU-Pro	0.7	+1.2 [-1.8, +4.2]	+1.5 [-1.7, +4.7]	+1.0 [-2.5, +4.7]	+0.8 [-2.7, +4.2]	+1.7 [-1.3, +4.8]	-0.3 [-3.5, +2.8]
		1.0	+1.3 [-2.5, +5.0]	+0.8 [-2.3, +4.2]	+2.0 [-1.5, +5.3]	+1.5 [-2.0, +4.7]	+1.2 [-2.3, +4.8]	+0.5 [-3.3, +4.2]
		1.3	+1.8 [-2.8, +6.7]	-0.3 [-4.2, +3.5]	+0.8 [-3.8, +5.5]	+0.8 [-3.5, +5.2]	-1.8 [-6.3, +2.7]	+3.7 [-0.8, +8.3]